

Group 50 Progress Report: Neural Network-Based Wildfire Risk Classification

Eric Solak, Tyler Yue, Ahmed Sahi
{solake, yuet5, sahia8}@mcmaster.ca

1 Introduction

Wildfires are a growing threat to infrastructure, ecosystems, and human safety. In addition to the immediate danger of fire, wildfire smoke can travel thousands of kilometers, degrading air quality and affecting millions of people [NASA \(2025b\)](#). Predicting the likelihood of wildfires in advance can help authorities, first responders, and communities with preparation, resource allocation, and reduce potential damage.

This project aims to classify regions into low, medium, and high wildfire risk using a machine learning model trained on environmental and meteorological data. The resulting risk predictions can support proactive decision making and reduce the overall impact of wildfires on both people and the environment.

1.1 Related Work

Predicting wildfires is a sought after problem in both environmental science and machine learning. Existing approaches can be categorized into deterministic models, statistical models, and machine learning models.

Deterministic models, such as the Canadian Forest Fire Danger Rating System (CFFDRS) rely on empirical formulae derived from field experiments to estimate wildfire risk based on meteorological factors [Government of Canada \(2025\)](#). These models are generally used in operational settings but are limited in their ability to capture non-linear interactions between variables.

Statistical models include regression-based and Bayesian approaches that predict wildfire likelihood using historical fire and weather data [Cisneros et al. \(2024\)](#); [Joseph et al. \(2019\)](#). These methods are effective in identifying trends but they are limited in their ability to capture rare or extreme weather conditions, which is critical for high-risk classification.

Machine learning approaches have used tree-ensemble methods such as XGBoost to classify wildfire risk based on tabular environmental data [Hu et al. \(2025\)](#). More recently however, neural networks have been trained on both satellite imagery and meteorological data to predict wildfire spread [Shadrin et al. \(2024\)](#). These models can learn complex, non-linear relationships but require preprocessing and normalization of inputs.

2 Dataset

For this project, we will utilize a Wildfire Prediction Dataset found on Kaggle [Vijayaragul \(2025\)](#). This dataset contains 118858 entries and 17 attributes, with no missing values. Further, all values are numerical (float64), simplifying any data manipulation by avoiding issues revolving around categorical data.

2.1 Data Sources

There are two main sources used by this dataset: NASA's FIRMS VIIRS SCC system for fire radiative power (FRP) data, and Open-Meteo's meteorological data [NASA \(2025a\)](#); [Open-Meteo \(2025\)](#). As these sources are open and publicly available, the data used is compliant with all terms of service.

2.2 Features

The dataset contains the following features:

- **daynight_N**: Indicator for whether the observation was taken during the day or night.
- **lat, lon**: Latitude and longitude coordinates of the location where the observation was recorded.
- **fire_weather_index**: An index representing the overall fire risk based on meteorological conditions.
- **pressure_mean**: Mean atmospheric pressure at the location.

- **wind_direction_mean, wind_direction_std:** Mean and standard deviation of wind direction.
- **solar_radiation_mean:** Average solar radiation received.
- **dewpoint_mean:** Average dew point temperature.
- **cloud_cover_mean:** Average cloud cover percentage.
- **evapotranspiration_total:** Total evapotranspiration over the observation period.
- **humidity_min:** Minimum humidity recorded.
- **temp_mean, temp_range:** Mean temperature and temperature range.
- **wind_speed_max:** Maximum wind speed observed.
- **occured:** Binary flag indicating whether a fire occurred (1) or not (0).
- **frp:** Fire Radiative Power, measuring the intensity of a fire if it occurred.

2.3 Preprocessing

The dataset requires minimal preprocessing due to its numerical format. The preprocessing steps used are:

- **Clipping negative values:** Negative *fire_weather_index* values are set to 0, as negative FWI is meaningless (i.e., negative risk is nonsensical).
- **Feature normalization:** All numeric features are standardized to facilitate stable training of the neural network. Each feature was scaled to have a mean of 0 and a standard deviation of 1, ensuring that features with larger ranges do not dominate the training process.

2.4 Data Split

A standard training/validation/testing split will be used, 70% for training, 15% for validation, 15% for testing.

3 Features

All 17 numerical features are used as input to the neural network. These features, described in Section 2.2, include positional, meteorological and environmental data. Positional features (*lat*, *lon*) provide the geographical context, important to the input as wildfire risk is often region-specific due to climate and vegetation. Meteorological features, such as *temp_mean*, *temp_range*, *humidity_min*, *pressure_mean*, *cloud_cover_mean*, *dewpoint_mean*, *solar_radiation_mean*, etc, describe the atmospheric conditions that directly influence fire ignition and propagation. Environmental features, including *evapotranspiration_total*, *fire_weather_index*, etc, describe the conditions that influence fire behaviour and ignition potential.

The model's target variable is the wildfire risk category, derived from the *fire_weather_index* and *occurred* variables in the dataset, classified into low, medium, or high risk. By combining multiple features that capture positional, meteorological, and environmental conditions, the neural network can learn complex, non-linear relationships that contribute to wildfire risk.

4 Implementation

The neural network, implemented in `02_model_training.ipynb`, uses a feed-forward MLP optimized for meteorological data. The model takes 16 input features and predicts the probability of belonging to one of the three wildfire risk classes. A combination of linear, ReLU, softmax and dropout functions were used in this model. Model training ran for up to 95 epochs with early stopping (patience = 5).

The final model implemented is a Neural Network consisting of 4 fully connected layers. The model achieved an accuracy of 0.9448 on our test set. The architecture is as follows:

- Input Layer: 16 Features
- Hidden Layer 1: 256 neurons
- Hidden Layer 2: 128 neurons
- Hidden Layer 3: 64 neurons
- Output Layer: 3 neurons

The activation function used is ReLU.

Regularization consisted of the following methods:

- L2 Regularization using weight decay of strength 1×10^{-4}
- Batch Normalization after each hidden layer
- Dropout on each hidden layer with a dropout rate of 0.3 for the first two layers and 0.2 for the last hidden layer

Comparing our model with a simple baseline of a majority vote strategy we achieved much higher accuracy. Since there are 3 classes and low is the most common the majority vote strategy entails voting for the ‘Low’ class each time. This results in a test accuracy of 0.4998. Our model significantly outperformed this benchmark.

The model utilized a cross entropy loss function. Since the number of samples in each class was uneven we applied a weighted cross entropy loss to ensure that minority classes have a higher weight during training to prevent the model from being biased towards the majority class. This is shown in Figure 1.

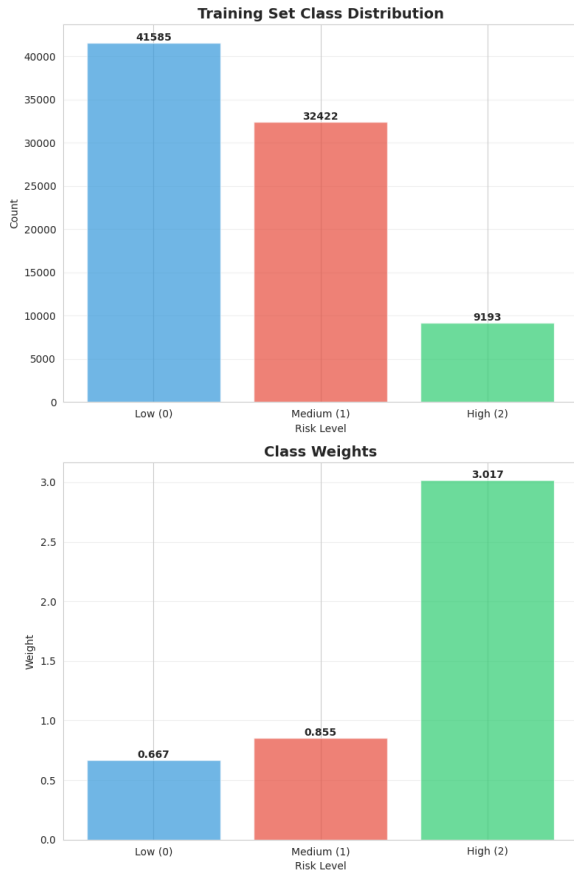
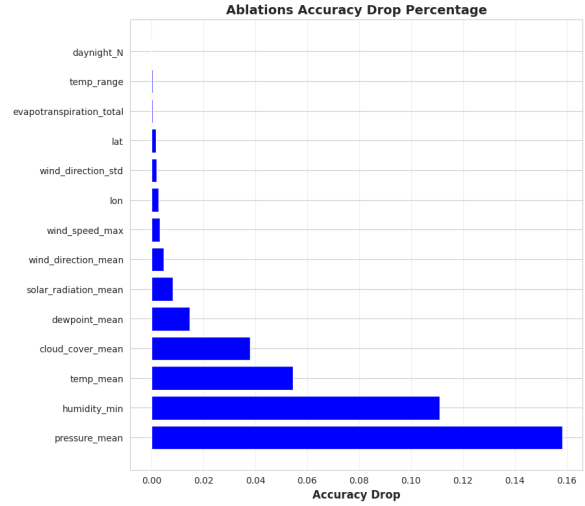


Figure 1: Weighted cross entropy loss applied to compensate for class imbalance.

The optimization function used is the AdamW optimizer. We used this optimization function over a traditional SGD optimizer since it experienced

faster convergence due to its parameter updating methodology which takes into account the moving averages of the mean and variance of the gradient as opposed to the SGD optimizer which updates based on the gradient of the loss.

The results of holding out specific features in the model is shown in Figure 2. This chart shows the effect on the accuracy of the model when holding out each specific feature. This shows that the most important features are pressure_mean and humidity_min.



Model accuracy when holding out individual features during training.

5 Results and Evaluation

Our model performs a three-way data split into training, test, and validation datasets. We used a stratified split to maintain the distribution of our target variable *risk_level* between the sets. The purpose of the splits are as follows:

1. **Training set:** used to fit the model parameters and to update the weights during backpropagation.
2. **Validation set:** used to determine performance during training and used for our early stop gradient descent and dynamic learning rate scheduling.
3. **Test set:** used to evaluate the model performance on unseen data and provide an accurate measurement of our model’s ability to generalize and predict on new data.

Our evaluation metrics are based on the classification performance. We computed the precision, recall, and F1-score for each risk class, together with the overall accuracy and a confusion matrix.

Accuracy serves as the primary evaluation metric, while the F1-score provides a more balanced measure of performance across all risk categories.

The neural-network classifier was trained on GPU for 95 epochs with early stopping. The model achieved a validation accuracy of 0.9439 before convergence and a test accuracy of 0.9432. The F1 scores show relatively good performance:

- **Low-risk (class 0):** F1 = 0.96
- **Medium-risk (class 1):** F1 = 0.93
- **High-risk (class 2):** F1 = 0.92

The confusion matrix shows that most misclassifications are between medium and high-risk wildfires. Precision and recall vary for the each class in the range of 0.94 to 1.00. Our metrics from the previous progress report were adequate, and with our modifications to the neural network, we achieved much greater accuracy and F1 scores.

6 Feedback and Plans

During the project, our TA provided guidance on narrowing the scope of our wildfire risk classification task and ensuring that our model design aligns with the project requirements. In particular, we were advised to focus on using the numerical environmental and meteorological data effectively, rather than attempting to include complex satellite imagery or external data sources, which could overcomplicate the project given the time constraints. This advice helped us narrow the project scope and prioritize the features most relevant to predicting wildfire risk. The TA also emphasized the importance of properly preprocessing features, specifically the negative and extreme values in *fire_weather_index* to ensure stable neural network training.

Based on this feedback and reflections on the project's progress so far, our plan for the remaining weeks is to finalize the preprocessing pipeline to ensure all unrealistic values are handled appropriately (extreme *fire_weather_index* values). We also plan on refining the presentation of the results and conduct exploratory analysis. This includes generating detailed visualizations of feature importance using partial dependence plots to explain the model's predictions for specific regions. Including these graphs in the final report will help communicate both the strengths and limitations of the model.

Team Contributions

- **Eric Solak:** Created Team Contract. Authored and completed Project Proposal sections *Overview*, and *Task Definition*. Set up the GitHub repository, including modularization, project structuring, separation of concerns. Authored and completed Project Progress Report sections 1 *Introduction*, 2 *Related Work*, 3 *Dataset*, 4 *Features*, and 7 *Feedback and Plans*. Implemented final neural network architecture and fixed data leakage to achieve 90% accuracy in final model.
- **Tyler Yue:** Authored and completed Project Proposal section *Results and Evaluation*. Coded parts of the model including the selection of layers in the neural network and adding a class weighting to improve accuracy by modifying the loss penalty for classes with less datapoints. Completed sections 4 and sections 5 of the report and updated to meet new model specifications and results. Tuned parameters of the model to increase accuracy including adjusting the epochs patience and activation functions. Created diagrams for inclusion in report.
- **Ahmed Sahi:** Coded the preprocessing steps in notebook 1, the model setup in notebook 2, and the evaluation metrics in notebook 3. Authored and completed Project Proposal section *Implementation*.

References

- Daniela Cisneros, Jordan Richards, Ashok Dahal, Luigi Lombardo, and Raphaël Huser. 2024. [Deep statistical regression for jointly moderate and extreme australian wildfires](#). *Spatial Statistics*.
- Government of Canada. 2025. [Canadian forest fire danger rating system](#).
- Chenglei Hu, Regina Baltazar Bispo, Håvard Rue, Carlos C. DaCamara, Ben Swallow, and Daniela Castro-Camilo. 2025. [Xgboost meets inla: a two-stage spatio-temporal forecasting of wildfires in portugal](#). *Cornell University*.
- Maxwell B Joseph, Matthew W Rossi, Nathan P Mitekiewicz, Adam L Mahood, Megan E Cattau, Lise Ann St Denis, R Chelsea Nagy, Virginia Iglesias, John T Abatzoglou, and Jennifer K Balch. 2019. [Deep statistical regression for jointly moderate and extreme australian wildfires](#). *National Center for Biotechnology Information*.

NASA. 2025a. [Fire information for resource management system](#).

NASA. 2025b. [Wildfires and climate change](#). *NASA Science*.

Open-Meteo. 2025. [Free weather api](#).

Dmitrii Shadrin, Svetlana Illarionova, Fedor Gubanov, Ksenia Evteeva, Maksim Mironenko, Ivan Levchunets, Roman Belousov, and Evgeny Burnaev. 2024. [Wildfire spreading prediction using multimodal data and deep neural network approach](#). *Scientific Reports*.

V. R. Vijayaragul. 2025. [Global wildfire dataset](#).